PRM323 — Lab 2: AI-Assisted UI/UX Design (Google Stitch → Figma)

In this lab you design a mobile app UI/UX with AI tools, and you are graded on your decisions as much as on the final screens. The process (analysis, prompting, critique, reasoning) is worth 45 points; the product (UI, design system, prototype, handoff) is worth 55.

1. Lab information

Item	Detail
Course	PRM323 — Cross-Platform Application Development
Lab	Lab 2 — AI-Assisted UI/UX Design
Work mode	Individual (unless the lecturer announces teams)
Duration / deadline	[to be filled by lecturer]
Weight in course grade	[to be filled by lecturer]
Required tools	Google Stitch, Figma (free Education plan), GitHub, one AI assistant of your choice for critique
Submission	One GitHub repository link and one Figma link (view access for anyone with the link)
Coding	None. No Flutter code is written in this lab

Tools: Google Stitch · Figma for students and educators

2. Learning outcomes

After this lab you can:

1. Define a target user, a problem and an information architecture before opening any design tool.

2. Write prompts that give Google Stitch enough context to produce usable UI, and refine them over several iterations.
3. Critique AI output against usability heuristics, accessibility criteria and your user, then accept, modify or reject each recommendation with a reason.
4. Build a token-based design system and reusable components with states in Figma.
5. Write a specification that another developer can implement in Flutter without asking you questions.

3. Task

Design a mobile application that solves a real problem for a specific group of users. Your final UI must contain:

- At least 8 distinct screens (dialogs and state variants do not count as separate screens).
- At least 3 complete user flows, each with a defined start, a goal and an end.
- At least 1 flow that includes an error or recovery path (for example, failed input, no network, or undo).

Topic. Choose from the list below or propose your own, and get it approved by the lecturer by [date]. Make it specific: write "a study planner for first-year students who have 6 courses and no fixed routine", not "a study planner".

- Student Course Planner (timetable and course selection)
- Study Planner (daily study sessions and deadlines)
- Campus Event App
- Expense Tracker
- Food Ordering App
- Habit Tracker
- Library App
- Internship Finder

4. Required workflow

Follow the stages in this order: Analyze, Generate, Critique, Refine, Prototype, Handoff. Each stage produces a deliverable that the next stage depends on.

#	Stage	Output
1	Problem and persona	`ux/persona.md`
2	UX analysis, information architecture, user flows	`ux/user-flow.md`
3	Design brief (color, type, spacing, tone, component rules)	`design/DESIGN.md`
4	Initial UI in Google Stitch	Prompt and screenshots in `ai/ai-design-log.md`
5	At least 3 AI iterations with AI critique	Log entries with decisions
6	Wireframe and final UI in Figma	Figma pages 02 and 03
7	Design system and components	Figma pages 04 and 05
8	Interactive prototype	Figma page 06
9	Flutter handoff	`handoff/flutter-handoff.md`

Tool note. Google Stitch features, quotas and export options change over time. If Stitch is unavailable to you, tell the lecturer before starting; an equivalent AI UI generator may be approved, but the log requirements stay the same. Screens exported from an AI tool usually arrive as loose layers, so plan to rebuild them in Figma with Auto Layout, tokens and components rather than relying on the import.

References

- Google Stitch
- Figma for students and educators
- Figma: Guide to auto layout

5. AI requirements

You must use AI, and you must record how. AI generates; you decide.

1. **Initial UI.** Generate the first version of your UI in Google Stitch. Record the full prompt, word for word, and save screenshots of the output.

2. **Three meaningful iterations.** An iteration counts only if it targets a specific, named problem (for example: primary action unclear, navigation too deep, low contrast, missing empty state) and differs from the other iterations. Save before and after screenshots. Asking for "make it nicer" does not count.
3. **AI critique.** Ask an AI assistant to review your UI against Nielsen's 10 usability heuristics, basic accessibility rules and your persona. Keep the prompt and the full answer. The critique must contain at least 5 findings tied to specific screens.
4. **Decision record.** For every important recommendation, write whether you accepted, modified or rejected it, and why. The reason must refer to your persona, a heuristic, or a real constraint (scope, time, platform). "I liked it" and "the AI said so" are not reasons.
5. **Disclosure.** Log every AI tool you used, including tools outside Stitch, and for what purpose.

References

- Nielsen Norman Group: 10 Usability Heuristics for User Interface Design
- W3C: Understanding WCAG 2.1 Success Criterion 1.4.3, Contrast (Minimum)
- Google Stitch

6. UX analysis requirements

`ux/persona.md` must contain:

- One primary persona: name, age range, role, daily context, goals, pain points, device habits, and any accessibility needs.
- A one-sentence problem statement.
- One measurable success criterion (for example: "a student can add a new deadline in under 20 seconds").

`ux/user-flow.md` must contain:

- Information architecture: a screen inventory and the navigation structure (which screens are top-level, which are nested).
- At least 3 user flows, each drawn as a diagram (Mermaid is fine) with a start, a goal, an end, the happy path and at least one alternative path. One flow must include an error or recovery path.
- A table mapping each flow to the screens it uses.

References

- Nielsen Norman Group: Personas Make Users Memorable for Product Team Members
- Mermaid: Flowchart syntax

7. Figma requirements

The Figma file must have exactly these six pages, in this order:

Page	Content
01 User Flow	Your flow diagrams, each linked to the screens it uses
02 Wireframe	Low-fidelity layout of all screens
03 Final UI	All final screens, at 360 × 800 dp, with state variants placed next to the screens they belong to
04 Design System	Tokens: color, typography, spacing, corner radius, elevation
05 Components	Reusable component library with variants
06 Prototype	Clickable flows with a defined starting point for each

Design system. Define tokens as Figma Variables or styles. Components must use the tokens; hard-coded colors or sizes inside components are penalized. A light theme is required; a dark theme is optional.

Minimum components, each built with Auto Layout and variants:

Component	Required states
Button	Default, pressed, disabled, loading
Text field	Default, focused, filled, error, disabled
Card	Default, pressed (if tappable)
Navigation (bottom bar or equivalent)	Each destination selected
App bar	Default, with actions
Dialog	Confirmation, and one destructive or error variant

Component	Required states
Loading state	At least one screen-level pattern (skeleton or spinner)
Empty state	Illustration or icon, message, one action
Error state	Message, cause in plain language, retry action

Accessibility and responsive checklist. Record the result of each item in `design/design-decisions.md`.

- Text contrast of at least 4.5:1 for body text and 3:1 for large text and UI controls (WCAG 2.x).
- Touch targets of at least 48 × 48 dp.
- Body text of at least 14 sp; no information conveyed by color alone.
- Layout uses constraints and Auto Layout, and is checked at 360 dp and 412 dp widths.
- In `handoff/flutter-handoff.md`, describe what changes on wider screens (600 dp and above). You do not need to design them.

Prototype. Each of your 3 flows must be clickable from start to end without dead ends. Include at least one dialog opened as an overlay and one loading-to-result transition. Set a flow starting point for each flow in Figma.

References

- Figma: Guide to variables
- Figma: The difference between variables and styles
- Figma: Guide to auto layout
- Figma: Create and use variants
- Figma: Create and manage component properties
- Figma: Guide to prototyping
- Figma: Create and manage prototype flows
- W3C: Understanding WCAG 2.1 Success Criterion 1.4.3, Contrast (Minimum)
- Material Design 3: Accessibility designing
- Flutter: Accessibility, UI design and styling

8. Repository and Flutter handoff

Submit one GitHub repository named `prm323-lab2-<student-id>` with this structure:

```
prm323-lab2-<student-id>/
├── README.md
├── ux/
|   ├── persona.md
|   └── user-flow.md
├── design/
|   ├── DESIGN.md
|   ├── screen-spec.md
|   └── design-decisions.md
├── ai/
|   └── ai-design-log.md
├── assets/
|   ├── stitch/          (screenshots of every generated version)
|   └── figma/           (exported PNGs of final screens)
└── handoff/
    └── flutter-handoff.md
```

File	Purpose
`README.md`	Topic, one-line persona, Figma link, list of AI tools used, and a map of where to find each deliverable
`design/DESIGN.md`	The design brief you gave to Stitch: color, typography, spacing, tone, component rules
`design/screen-spec.md`	What each screen is for, what content it shows and which flow it belongs to (design intent)
`design/design-decisions.md`	The 10 or fewer most important decisions with reasons, plus the accessibility checklist results
`ai/ai-design-log.md`	Every prompt, output, critique and decision (template in Appendix A)
`handoff/flutter-handoff.md`	How a developer builds the app (below)

Flutter handoff. For every screen, specify all six items:

1. **Layout:** the structure from top to bottom (for example `Scaffold` → `AppBar` → scrollable `Column`), spacing values from your tokens, and what scrolls.

2. **Components:** which components from your library appear, with variant names.
3. **States:** default, loading, empty, error and populated, whichever apply.
4. **User interactions:** each tap, swipe, input and validation rule, and what it causes.
5. **Navigation:** where the user comes from, where each action leads, and what Back does.
6. **Important UI constraints:** minimum touch target, keyboard behavior, text overflow rules, maximum content lengths, behavior on wider screens.

Also include two global tables: **tokens to Flutter** (each color, text style and spacing value with its intended `ColorScheme`, `TextTheme` or constant name) and **components to widgets** (for example Button to `FilledButton`, Text field to `TextFormField`, Navigation to `NavigationBar`, Dialog to `AlertDialog`). A template is in Appendix B.

Handoff test. A classmate should be able to build one screen from your handoff, using Figma only to look at visuals. Anything they have to ask you is a gap in the specification.

References

- Flutter: Use themes to share colors and font styles
- Flutter API: ThemeData
- Flutter: Material components (widget catalog)
- Flutter: Navigation and routing
- Flutter: Build a form with validation
- Flutter: Adaptive and responsive design

9. Grading rubric (100 points)

Each sub-criterion is graded as full marks, partial marks or zero. Partial marks are given when the requirement is met in part, for example on some flows or screens but not all. Process criteria total 45 points (UX 15, AI generation 10, iteration and critique 15, documentation 5); product criteria total 55.

Criterion	Pts	Full marks when	Points are lost when
UX analysis and user flow	**15**		

Criterion	Pts	Full marks when	Points are lost when
Target user and problem	5	Persona is specific and has context, goals and pain points; one-sentence problem; measurable success criterion	Persona is generic ("students"); no pain points; no success criterion
Information architecture	5	Screen inventory and navigation structure; every screen is used by at least one flow	Screens belong to no flow; screens cannot be reached; no navigation structure
User flows	5	At least 3 flows with start, goal, end, alternative path, and a flow-to-screen table	Only happy paths; fewer than 3 flows; flows not mapped to screens
AI-assisted UI generation	**10**		
Prompting	5	Prompt states user, task, platform, constraints and style (based on `DESIGN.md`) and is recorded verbatim	One-line prompt; prompt not recorded or summarized
Initial UI from Stitch	5	Generated UI covers at least the main flow, with screenshots as evidence	No evidence of Stitch output; output covers only one screen
AI iteration and UX critique	**15**		
At least 3 meaningful iterations	5	Each iteration targets a different named problem, with before and after screenshots	Fewer than 3; cosmetic rerolls; iterations repeat the same change
AI-based UX critique	5	Critique uses named heuristics, accessibility and persona; at least 5 findings tied to screens	Vague critique; fewer than 5 findings; findings not tied to screens

Criterion	Pts	Full marks when	Points are lost when
Student reasoning and decisions	5	Every important recommendation marked accepted, modified or rejected with a reason, and Figma reflects the decisions	No reasons; everything accepted without justification; decisions not visible in the final UI
Final UI/UX design	**20**		
Visual hierarchy	5	One clear primary action per screen; consistent type scale; grouping is obvious	Competing primary actions; inconsistent text sizes
Consistency	5	Same components, tokens, labels and patterns across all screens	Same element looks or is named differently on different screens
Usability	5	Flows completable with few steps; feedback for every action; error prevention and recovery; empty, loading and error states on relevant screens	Dead ends; no feedback; states missing
Accessibility and responsive	5	Checklist in section 7 completed with evidence; layout checked at 360 dp and 412 dp	Contrast or touch-target failures; color-only meaning; checklist missing
Design system and components	**15**		
Design tokens	5	Color, type, spacing, radius and elevation defined as Variables or styles and used by components	Hard-coded values; tokens defined but not used
Reusable components	5	All 9 minimum components built with Auto Layout and variants; screens use instances	Missing components; screens use detached copies

Criterion	Pts	Full marks when	Points are lost when
Component states	5	Every required state in section 7 exists and is used	States missing or only drawn as separate loose frames
Interactive prototype	**10**		
At least 3 complete flows	5	Each flow is clickable from start to end with a defined starting point	Fewer than 3 flows; flows break midway
Correct interactions and navigation	5	Back behavior is correct; at least one overlay and one loading-to-result transition; no dead ends	Wrong destinations; no Back; dead ends
Flutter handoff specification	**10**		
Screen specifications	5	Every screen has all 6 items from section 8	Screens missing; items missing; only descriptions of looks
Components, states, interactions, mappings	5	Token-to-Flutter and component-to-widget tables complete; states and interactions unambiguous	A developer would have to ask questions to build a screen
Documentation and AI usage log	**5**		
Complete prompts and iterations	3	All prompts, outputs, critique and decisions present and traceable to screenshots	Prompts summarized or missing; iterations without evidence
Clear documentation	2	Repository matches the structure; README maps deliverables; all links work	Wrong structure; broken or private links

Access rule. If your Figma link cannot be opened at the deadline, the Figma-based criteria (final UI, design system, prototype) are graded 0 until it is fixed, and the late policy applies.

10. Academic integrity, verification and submission

Using AI is required, not forbidden. What is not allowed is submitting AI output as your own understanding. To make the process checkable:

- Your Git history must show work over time, not one upload at the end (at least [N] commits over at least [N] days).
- Figma version history is reviewed; a file that appears complete in one save will be questioned.
- Each student may be asked for a short review of about 5 minutes: walk through one flow in the prototype and explain two decisions from your log. Marks in the process criteria can be adjusted after this review.
- Undisclosed AI use, fabricated log entries, or copying another student's work are handled under the university's academic integrity policy.

Submission checklist

- [ ] Topic approved by the lecturer
- [ ] Persona and problem statement written
- [ ] Information architecture and 3 flows drawn, with an error path
- [ ] `DESIGN.md` written before the first Stitch prompt
- [ ] Initial Stitch prompt and screenshots saved
- [ ] At least 3 iterations, each with before and after screenshots
- [ ] AI critique saved with at least 5 findings
- [ ] Every important recommendation marked accepted, modified or rejected, with a reason
- [ ] Figma file has the six pages in order, with tokens and all 9 components and their states
- [ ] Prototype runs 3 flows from start to end
- [ ] Accessibility checklist completed in `design-decisions.md`
- [ ] `flutter-handoff.md` covers every screen and both mapping tables
- [ ] README lists AI tools used and the Figma link
- [ ] Figma link opens in a private browser window